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Survival Analysis: Heart Failure Clinical Records

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1. Introduction

Heart failure is a serious cardiovascular condition affecting millions worldwide. This analysis applies survival analysis techniques to the **Heart Failure Clinical Records Dataset** (Chicco & Jurman, 2020) to identify clinical predictors of mortality.

Objectives:

- Estimate survival functions using Kaplan-Meier (KM) curves
- Compare survival between clinically relevant subgroups via log-rank tests
- Identify independent predictors of death using a Cox Proportional Hazards model

2. Data Overview

The dataset contains **299 patients** with heart failure, recruited at the Faisalabad Institute of Cardiology (Pakistan, 2015). The follow-up period ranged from 4 to 285 days.

Variable	Description	Type
age	Patient age (years)	Continuous
anaemia	Decrease in red blood cells	Binary
creatinine_phosphokinase	CPK enzyme level (mcg/L)	Continuous
diabetes	Presence of diabetes	Binary
ejection_fraction	% blood leaving heart per contraction	Continuous
high_blood_pressure	Presence of hypertension	Binary
platelets	Platelets in blood (kiloplatelets/mL)	Continuous
serum_creatinine	Serum creatinine level (mg/dL)	Continuous
serum_sodium	Serum sodium level (mEq/L)	Continuous
sex	Patient sex (1 = Male, 0 = Female)	Binary
smoking	Smoking status	Binary
time	Follow-up period (days)	Continuous
DEATH_EVENT	Death during follow-up (1 = Yes)	Binary (outcome)

3. Setup & Data Loading

4. Descriptive Statistics

Table 1. Patient Characteristics (N = 299)

Characteristic	N = 299
age	60.83 (11.89)
anaemia	129 (43%)
creatinine_phosphokinase	581.84 (970.29)
diabetes	125 (42%)
ejection_fraction	38.08 (11.83)
high_blood_pressure	105 (35%)
platelets	263,358.03 (97,804.24)
serum_creatinine	1.39 (1.03)
serum_sodium	136.63 (4.41)
sex	
Female	105 (35%)
Male	194 (65%)
smoking	96 (32%)
age_group	
< 60	162 (54%)
60–70	85 (28%)
> 70	52 (17%)

[1] Mean (SD); n (%)

Table 2. Event Summary

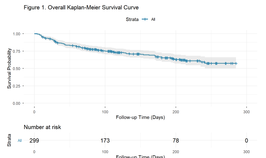
Total Patients	Deaths	Death Rate (%)	Median Follow-up (days)
299	96	32.1	115

Note: Median survival was not reached during the study period, as fewer than 50% of patients experienced the event (death rate = 32.1%).

5. Kaplan-Meier Survival Estimates

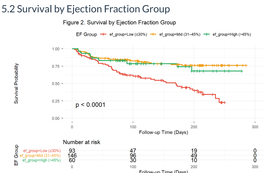
5.1 Overall Survival

Figure 1. Overall Kaplan-Meier Survival Curve



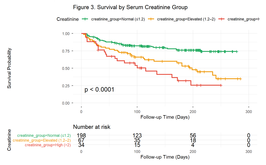
5.2 Survival by Ejection Fraction Group

Figure 2. Survival by Ejection Fraction Group



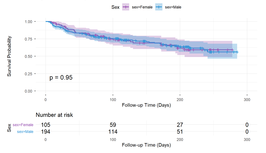
5.3 Survival by Serum Creatinine Group

Figure 3. Survival by Serum Creatinine Group



5.4 Survival by Sex

Figure 4. Survival by Sex



6. Log-Rank Tests

Table 3. Log-Rank Test Results

Variable	Chi_sq	df	p_value	Significant
Ejection Fraction Group	33.053	2	0.0000	✓
Serum Creatinine Group	33.474	2	0.0000	✓
Age Group	27.445	2	0.0000	✓
High Blood Pressure	4.406	1	0.0358	✓
Anaemia	2.726	1	0.0987	
Diabetes	0.041	1	0.8405	
Sex	0.004	1	0.9498	
Smoking	0.002	1	0.9640	

7. Cox Proportional Hazards Model

7.1 Univariable Cox Models

Table 4. Univariable Cox Regression Results

Variable	HR	95% CI	p_value
age	**1.043**	**(1.026 – 1.061)**	**<0.001**
ejection_fraction	**0.955**	**(0.935 – 0.975)**	**<0.001**
serum_creatinine	**1.336**	**(1.2 – 1.487)**	**<0.001**
serum_sodium	**0.935**	**(0.9 – 0.971)**	**<0.001**
high_blood_pressure	**1.546**	**(1.026 – 2.331)**	**0.0374**
anaemia	1.401	(0.938 – 2.094)	0.0998
creatinine_phosphokinase	1.000	(1 – 1)	0.2625
platelets	1.000	(1 – 1)	0.4659
diabetes	0.959	(0.639 – 1.44)	0.84
sex	1.014	(0.667 – 1.54)	0.9493
smoking	0.990	(0.643 – 1.525)	0.9653

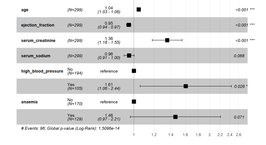
7.2 Multivariable Cox Model

Variables with $p < 0.10$ in univariable analysis are included.

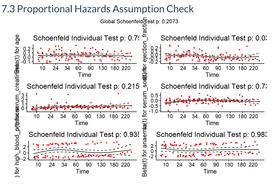
Table: Multivariable Cox Model Results

Variable	HR	95% CI Lower	95% CI Upper	p-value
Age	1.045	1.027	1.063	<0.001
Ejection Fraction	0.954	0.935	0.974	<0.001
Serum Creatinine	1.256	1.182	1.554	<0.001
Serum Sodium	0.958	0.914	1.003	0.068
High Blood Pressure (Yes)	1.606	1.058	2.437	0.026
Anaemia (Yes)	1.461	0.968	2.206	0.071

Figure 5. Forest Plot — Multivariable Cox Model



7.3 Proportional Hazards Assumption Check



8. Key Findings

Table 5. Summary of Key Findings

Finding	Statistical Support
32.1% mortality rate over a median follow-up of 115 days; median survival not reached	96 deaths among 299 patients; KM overall curve
Low ejection fraction (≤30%) significantly worsens survival	Log-rank p < 0.0001; HR = 0.954 per unit increase in multivariable Cox
Elevated serum creatinine is the strongest independent predictor of mortality	Log-rank p < 0.0001; HR = 1.256, p < 0.001 in multivariable Cox
Older age independently increases hazard of death	HR = 1.045 per year, p < 0.001 in multivariable Cox
High blood pressure independently associated with increased mortality risk	HR = 1.606, p = 0.026 in multivariable Cox
Serum sodium shows a marginal inverse association with hazard (p = 0.068)	HR = 0.958, p = 0.068; CI crosses 1.0

9. Conclusion

This survival analysis identified **age**, **ejection fraction**, **serum creatinine**, and **high blood pressure** as significant independent predictors of mortality in heart failure patients (all $p < 0.05$), with serum sodium showing a borderline association ($p = 0.068$). Serum creatinine carried the largest effect size (HR = 1.26), consistent with Chicco & Jurman (2020). The global Schoenfeld test confirmed the proportional hazards assumption was met ($p = 0.207$), though ejection fraction showed a marginal individual violation ($p = 0.033$) warranting cautious interpretation. Overall model discrimination was adequate (concordance = 0.70), supporting the clinical utility of these variables in predicting heart failure mortality.

10. References

- Chicco, D., & Jurman, G. (2020). Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Medical Informatics and Decision Making, 20*(16).
- Kaggle Dataset: Heart Failure Clinical Records — https://www.kaggle.com/datasets/andrewmvd/heart-failure-clinical-data